[INCH-POUND]
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SUPERSEDING
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FEDERAL SPECIFICATION

CRUSHER, JAW, PORTABLE, PRIMARY AGGREGATE PLANT, MOBILE, 75-TON PER HOUR CAPACITY

The General Services Administration has authorized the use of this specification by all Federal agencies.

1. SCOPE

1.1 Scope. This specification covers a complete primary, portable, mobile, all electric powered plant capable of crushing aggregate at a production rate of 75 tons per hour (tph) (68 metric tons per hour (mtph)). The plant will hereinafter be called the unit.

1.2 Classification.

1.2.1 Type. This specification covers two types of crushers as specified (see 6.2).

Type I - Externally powered crusher
Type II - Crusher with self contained power plant

2. APPLICABLE DOCUMENTS

2.1 Government publications. The following documents, of the issues in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data which may improve this document should be sent to: Commanding Officer (Code 15E2), Naval Construction Battalion Center, 1000 23rd Avenue, Port Hueneme, CA 93043-4301, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Federal Standard

FED-STD-595 - Colors Used in Government Procurement.

Military Standards

MIL-STD-209 - Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MS51336 - Lunette-Coupler, Drawbar, Ring.

(Copies of federal and military specifications and standards required by contractors in connection with specific procurement functions are obtained from the Defense Automated Printing Services, Attn: DoDSSP, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.1 Other Government documents and publications. The following other Government documents and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DEPARTMENT OF LABOR (DoL)
OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

Occupational Safety and Health Standards.

(Application for copies of DoL standards should be addressed to the Superintendent of Documents, Government Printing Office, Washington DC 20402-0001.)

DEPARTMENT OF TRANSPORTATION (DoT)

Federal Motor Vehicle Safety Standards (FMVSS).
Federal Motor Carrier Safety Regulations (FMCSR).

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specified issue in effect on the date of invitation for bids or request for proposal shall apply.

ASTM

ASTM C 131 - Test Method for Resistance to Degradation of Small-Size Coarse Aggregate by Abrasion and Impact in the Los Angeles Machine.
ASTM C 170 - Test Method for Compressive Strength of Dimension Stone.

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(Private sector and civil agencies may purchase copies of these voluntary standards from the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA AB1 - Molded Case Circuit Breakers.
NEMA ICS1 - General Standards for Industrial Control and Systems.
NEMA ICS6 - Enclosures for Industrial Controls and Systems.
NEMA MG1 - Motors and Generators.
NEMA PB2 - Deadfront Distribution Switchboards.

(Private sector and civil agencies may purchase copies of these voluntary standards from the National Electrical Manufacturers Association, 1300 North 17th Street, Suite 1847, Rosslyn, VA 22209.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 - Equipment, Electrical, Maintenance.

(Private sector and civil agencies may purchase copies of these voluntary standards from the National Fire Protection Association, Battery March Park, Quincy, MA 02269.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE J534 - Lubrication Fittings.

(Private sector and civil agencies may purchase copies of these voluntary standards from the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

TIRE AND RIM ASSOCIATION, INC. (TRA)

TRA Yearbook.

(Private sector and civil agencies may purchase copies of these voluntary standards from the Tire and Rim Association, Inc., 175 Montrose West Avenue, Suite 150, Copley, OH 44321.)

(DoD activities may obtain copies of those adopted voluntary standards listed in the DoD Index of Specifications and Standards free of charge from the Defense Automated Printing Services, Attn: DoDSSP, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Description. The type I unit shall consist essentially of a hopper, feeder, jaw crusher, conveyor system, electric power system, operator's platform, and all accessories for a complete portable unit. The type II unit will consist of a diesel electric generator, a hopper, feeder, jaw crusher, conveyor system, electric power system, operator's platform, and all accessories for a complete portable unit. The unit shall be mounted on a pneumatic-tired semitrailer equipped with fifth wheel pin, applicable to any standard truck tractor, and a pneumatic-tired towing front dolly.

3.2 Standard commercial product. The crusher shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the crusher being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.3 First article. When specified (see 6.2), the contractor shall furnish a crusher for first article inspection and approval (see 4.2.1 and 6.3).

3.4 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. Unless otherwise specified, none of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification.

Prohibited materials. Asbestos, cadmium, and radioactive material shall not be used in this item. Radioactive material as defined by Title 10, Code of Federal Regulations (CFR), Parts 30 and 40, and other radioactive material in which the specific activity is greater than 0.002 microcuries per gram or activity per item equals or exceeds 0.01 microcuries.

3.5 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to ensure interchangeability of component parts, assemblies, and spare parts.

3.6 Maintainability. The unit shall be so designed and constructed that normal adjustment, repair, and overhaul can be readily accomplished by means of general purpose tools with a minimum removal or disturbance of other elements of the unit. Ease of maintenance provisions shall ensure operating clearances for facilitating maintenance and servicing. When specified (see 6.2), major components removed for maintenance shall have lifting eyes attached such as the stationary and

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movable jaws. Where possible, intricate locking devices, controls, and threaded fastenings, that can be easily over-torqued by operators lacking feel through thick gloves or numbness, shall be avoided. Covers or plates that must be removed for component adjustment or for parts removal shall be equipped with substantial fastenings.

3.7 Safety. All rotating or reciprocating parts, and all parts subject to high operational temperature, that are of such nature or are so located as to be or become a hazard to operating personnel, shall be adequately guarded or insulated to the extent necessary to eliminate the hazard. All platforms, steps, and handholds shall be provided for easy access to the operator's position and other work areas. The unit shall comply with OSHA regulations in effect at time of manufacture.

3.7.1 Noise levels. The noise levels shall be less than 85 dB(A) at the operator's position and other occasionally occupied positions as measured during typical operating conditioned and environments. If noise level cannot be reduced to below 85 dB(A) then hearing protection warnings shall be placed on the unit where they can be read by those personnel exposed to the hazard.

3.7.2 Dust suppression system. The equipment shall be provided with a dust suppression system to comply with OSHA regulations applicable at the time of manufacture.

3.8 Performance. The unit shall be capable of crushing any variety of rock having an average compressive strength of not less than 32,000 pounds per square inch (psi) (220 632 kilopascal (kPa)) when tested in accordance with ASTM C 170, having a percentage of wear not greater than 20 when tested in accordance with ASTM C 131, and a free moisture content of not greater than 6.5 percent. The unit shall be capable of crushing rock of all sizes including quarry rock up to 18 by 28 inches (457 millimetre (mm) by 711 mm) cross-section and a length receptacle to the primary jaw crusher opening depth (approximately 38 inches (965 mm)). The unit shall be capable of producing, with the crusher setting (close of stroke) set as follows:

Crusher Setting	Minimum Production
3 inches (76 mm)	75 tph (68 mtph)
4 inches (101 mm)	100 tph (91 mtph)
5 inches (127 mm)	120 tph (109 mtph)
6 inches (152 mm)	140 tph (127 mtph)

Provisions shall be made for an optional belt reject conveyor to remove all or a portion of the fines.

3.9 Feeder. The unit shall be provided with an apron feeder, wobbler feeder, vibrating grizzly feeder, or a combination thereof as specified (see 6.2). The feeder shall be capable of delivering up to 180 tph (163 mtph). When the unit incorporates a wobbler feeder or vibrating grizzly feeder, the bar spacing shall be adjustable from not less than 3 to 6 inches (76 to 152 mm). Replaceable liners, flights, bars, or plates shall be fabricated of alloy steel or high quality abrasive-resisting steel plate. When feeder incorporates a wobbler or vibrating grizzly, the feeder shall

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include a bypass spout and reject spout with gate to the reject conveyor. Protection shall be provided against shock and vibration being transmitted to the unit structures. The unit shall be controlled from the operator's platform.

3.10 Loading hopper. A loading hopper extending not less than the full length of the feeder with a vertical height above the feeder pan of not less than 24 inches (610 mm) shall be provided. The loading hopper shall break down into three sections (two sides and a rear). The feeder or feeder support and hopper shall be removable to reduce cube during transport. The sides and rear shall be mounted at an angle of not less than 30 degrees from the horizontal. The loading hopper shall be made with supports and brackets for wider extension. The top edge of the loading hopper shall be structurally reinforced to receive and support an additional wobbler or grizzly if required at a later date.

3.11 By-pass chute. A by-pass chute shall be provided to allow material smaller than the closed jaw setting to by-pass the jaw crusher. The chute shall be located between the feeder and the jaw crusher. The wobbler or grizzly shall be mounted across the chute opening to route material into the jaw crusher. The chute shall be protected at the point of impact of material by suitable replaceable liners with access for replacement.

3.12 Jaw crusher. The jaw crusher for the unit shall be V-belt driven from the unit motive power, and shall be of the overhead eccentric single jaw type with a jaw size opening of not less than 20 by 36 inches (508 by 914 mm). The jaws shall be approximately 40 inches (1 016 mm) in depth with the lower jaw opening adjustable from 2.5 to 6 inches (64 to 152 mm) in the closed position. The jaw crusher shall be of the hydraulic-shim-adjusted type with reversible stationary and removable jaw plates with self-aligning antifriction or tapered roller bearings. Jaw plates shall be equipped with toggle plates or safety relief plates for prevention of overload damage by crushable material (tramp iron). The eccentric shaft shall be fitted with counterweighted flywheels with sufficient mass to maintain relatively constant rotational speed throughout each revolution when crushing rock of maximum size and hardness. An extra set of cheek and jaw plates with appropriate toggle plates shall be furnished with the unit.

3.13 Conveyors. Conveyors shall be used to transport material within the unit and shall be of the belted type in accordance with the manufacturer's standard practice, providing a smooth and efficient flow of material. A conveyor shall be provided to transfer material from the jaw crusher and by-pass chute to the main product conveyor. The main product conveyor shall be capable of delivering 180 tph (163 mtph), discharging the product not less than 8 feet (2 438 mm) from the trailer frame and not less than 8 feet (2 438 mm) above ground level. A conveyor shall be provided for transferring material from the reject spout.

3.13.1 Belt type conveyors. Belt-type conveyors shall be equipped with troughing type idlers or rollers with spacing not greater than 4 feet (1 219 mm) between centers. Troughing idlers shall be so designed as to ensure true running and self-centering of the conveyor belts. Special rolls designed to absorb shock shall be installed under the conveyor belt, below the receiving area. Return idlers shall be provided with a spacing of not greater than 10 feet (3 084 mm) between centers. All idlers shall be not less than 4 inches (102 mm) in diameter. The width and running

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speed of all conveyor belts shall be such as to assure efficient movement of material, with a minimum of spillage and pileup at receiving points when the unit is operating at full capacity. An accessible adjustment for maintaining proper belt tension shall be provided. Cleaning devices, capable of removing foreign material adhering to the pulley side of the belt, shall be provided. Conveyors which extend outside the basic dimensional limits of the unit shall be of the folding, retractable, or readily demountable type. Rubber coating on the carrying side of the belt shall be not less than 0.1875-inch (5 mm) thick and rubber coating on the bottom side shall be not less than 0.0625-inch (1.6 mm) thick. Each belt-type conveyor shall be driven from a separate electric or hydraulic motor. The capacity of the motors shall be sufficient to handle the loads encountered when the plant is operating at maximum capacity. Motors shall be positioned to transmit power directly or by V-belt to the head pulley. Inclined conveyors shall be designed to prevent rollback of the belt in the event of power failure.

3.14 Operator's platform. The operator's platform shall be centrally located to enable the operator to observe the unit feeder, conveyor, and crusher operation. Controls for plant operation shall be mounted on the platform for safe convenient operation of the unit by the operator. Access ladders, handrails, and guardrails shall be provided. Antiskid surfaces shall be provided for the deck, catwalks, and ladders. Platforms and catwalks shall be furnished to provide access to points requiring routine maintenance and adjustment. All guardrails around the platform shall be demountable to reduce cube during transport.

3.15 Bearings. On this unit, all antifriction or bronze bearings exposed to the elements shall be provided with a dust-seal device to prevent the entry of abrasive material.

3.16 Electrical equipment. All electrical motors shall conform to NEMA MG1 polyphase induction, totally enclosed, fan cooled type. The rated horsepower (hp) of each motor shall be equal to, or greater than, the hp requirements of the equipment driven under all operating conditions. The hp of the motors shall be such as to allow progressive starting of all electrically operated components. The electric motors shall be suitable for operation on 460 volt (V), 60 hertz (Hz), 3-phase, alternating current from an outside source for type I units or the self contained generator for type II units. A main distribution panel conforming to NEMA PB2 shall provide circuit breakers for each electrical branch, and a main circuit breaker that will disconnect the entire unit from the electrical power supply. Circuit breakers shall conform to NEMA AB1. The main distribution panel shall be provided with a main feeder power receptacle, auxiliary main feeder outlet receptacle, and five auxiliary branch circuit receptacles mounted in a bank for receiving connections from portable conveyors. All power receptacles shall have spring-loaded covers. A motor controller operated by momentary contact start, and maintained contact stop pushbuttons shall be provided for control of each electric motor. Motor controllers shall conform to NEMA ICS1. Motor control pushbuttons and an emergency panic pushbutton, which controls tripping of the master circuit breaker, shall be installed in an easily accessible location at the operator's platform and at each control station for each electric motor used on the unit. Panic pushbutton enclosures shall be red. All enclosures for motors, motor controllers, distribution panel, pushbuttons, and other similar electrical apparatus shall be weatherproof and dustproof in accordance with NEMA ICS-6, type 4 enclosures. All wiring of the unit shall be furnished, including interconnections between individual components and between components and the

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distribution panel to provide an operable unit when connected to a 460V, 60 Hz, 3-phase power source. Type I units shall be furnished with a 100-foot (30 480 mm) main feeder cable. A permanently legible wiring diagram shall be installed on or near the main distribution panel. The function of all circuit breakers and controls shall be permanently marked on their respective panels or enclosures. Conductors shall be color coded or suitably identified. All wiring shall be adequately protected from physical damage. The electrical system, wiring practices, system protection, grounding, and wiring hardware shall meet the applicable requirements of NFPA 70. A grounding stud suitable for receiving a grounding electrode conductor of suitable size shall be provided for power cabinets, generators, and electric motors.

3.16.1 Diesel engine driven generator. Type II units shall be equipped with a complete, self-contained, diesel engine driven generator The generator shall contain all accessories necessary to power the unit for a period of not less than ten hours without servicing or refueling. The generator shall provide sufficient 460 volt power to allow the unit to operate under the maximum output conditions for the full ten-hour period.

3.17 Semitrailer. The maximum stress imposed on any structural component of the semitrailer shall be not greater than 60 percent of the yield point. The semitrailer for the crushing unit shall be equipped with a dolly and shall be capable of being safely towed over the road at speeds of not less than 15 miles per hour (mph) (24 kilometre per hour (km/h)). The dolly drawbar shall be equipped with a lunette ring in accordance with MS51336. The angle of departure shall be not less than 25 degrees and projections below the main frame members shall be held to a minimum. The gross weight on any one axle, with the unit prepared for travel, shall be not greater than 20,000 pounds (9 060 kilogram (kg)). The trailer shall comply with FMCSR and FMVSS requirements applicable to this item at the time of manufacture. The trailer shall have a minimum ground clearance of 12 inches (305 mm).

3.17.1 Brakes. The semitrailer shall be equipped with full service air service brakes controllable from the towing vehicle cab. Brake systems shall comply FMVSS 121. All necessary air hose and standard air couplings for connecting to the brake system of a towing vehicle shall be provided. The service brake installation shall include a breakaway system capable of automatically applying the brakes in case the semitrailer is accidentally separated from the towing vehicle, and shall be capable of maintaining the application for not less than 15 minutes. Parking brakes or chock blocks securely chained to the vehicle frame shall be provided for each wheel. Chock block chains shall be of sufficient length to permit placing the blocks either ahead or behind the wheels. The blocks shall be equipped with snaphooks for storage.

3.17.2 Jacks. Not more than six screw-jacks shall be provided with each unit. The jacks shall be capable of supporting the unit when fully equipped and fully loaded. Supports at each end of the unit and near the crusher jaw shall be provided for blocking during unit operation. The jacks shall be capable of elevating the king pin to a height of not less than 67 inches (1 702 mm) from the ground.

3.17.3 Lights. The trailer shall be equipped with a lighting system which shall comply with FMVSS 108.

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3.17.4 Tires and rims. The tire and rim ratings shall conform to TRA recommendations for the type and size of tires furnished. Tires shall be tubeless type with highway tread. Tires shall be of rated capacity not less than equal to the load imposed on each tire, measured at each wheel at the ground. Tires shall be not less than 100 level quality and shall be of domestic make. The recommended tire pressure shall be stenciled in 1 inch (25.4 mm) block letters above each tire or set of tires.

3.18 Identification plate. The contracting officer will furnish to the Government inspector the required identification plates. The contractor will be required to stamp the necessary data in the blank spaces thereon and securely affix said plates in a conspicuous place on each unit, assembly or subassembly, and parts as directed by the Government inspector. Nonferrous screws, rivets, or bolts of not less 0.125-inch (3 mm) in diameter shall be used to affix the plates. The applicable nomenclature contained in the contract item description shall be placed in the top blank.

3.19 Cleaning, treatment, and painting. Surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. The color of the finish coat conforming to FED-STD-595, shall as specified (see 6.2). Surfaces to be painted shall be cleaned and dried to ensure that they are free from contaminants such as oil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion product, or any other contaminating substances. As soon as practicable after cleaning, and before any corrosion product or other contamination can result, the surfaces shall be prepared or treated to ensure the adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat. The primer shall be applied over a properly pre-treated surface as soon as practicable. Painting shall be with manufacturer's current materials according to manufacturer's current processes and best commercial practice and the total dry film thickness shall be not less than 3.2 mils (0.081 3 mm) over the entire surface. The paint shall be free from runs, sags, orange peel, and other defects.

3.20 Instruction plates. The unit shall be equipped with instruction plates suitably located, describing any special or important procedures to be followed in operating and servicing the equipment. Plates shall be of a material which will last and remain legible for the life of the equipment, and shall be securely affixed thereto with nonferrous screws or bolts of not less than 0.125-inch (3 mm) diameter.

3.21 Toolbox(s). One or more weatherproof toolbox(es) shall be provided. The toolbox(es) shall be large enough to store all tools required for field service or maintenance. The toolbox(es) shall be mounted in a protected accessible location. The toolbox shall be located so that it may be reached from the ground without the aid of a ladder and shall have a locking hasp capable of accepting a lock with a 0.625-inch (16 mm) shackle.

3.22 Stenciling. The gross vehicle weight of each demountable component of the unit shall be stenciled on each side of the component in such a manner as to be readily discernible to personnel handling the equipment. The total gross weight of the unit shall be stamped on the identification plate by the contractor.

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3.23 Prime equipment accessories. Prime equipment shall be shipped with all required accessories and tools unless a written deviation is received from the contracting officer.

3.24 Lifting and tiedown attachments. The unit shall be equipped with lifting and tiedown attachments. Lifting and tiedown attachments shall conform to type II or type III of MIL-STD-209. A nonferrous transportation plate shall be provided and mechanically attached to the unit. Transportation plates shall be inscribed with a diagram showing the lifting attachments and lifting slings, the capacity of each attachment, and the required length and size of each sling cable. A silhouette of the item furnished showing the center of gravity shall be provided on the transportation plate. Tiedown attachments may be identified by stenciling or other suitable marking. Tiedown marking shall clearly indicate that the attachments are intended for the tying down of the unit on the carrier when shipped.

3.25 Lubrication. Means for lubrication shall be in accordance with the manufacturer's standard practice. The lubricating points shall be easily visible and accessible. Hydraulic lubrication fittings shall be in accordance with SAE J534. Where use of high-pressure lubricating equipment, 1,000 psi (6 895 kPa) or higher, will damage grease seals or other parts, a suitable warning shall be affixed to the equipment in a conspicuous location. The unit shall be lubricated as specified in section 5.

3.26 Workmanship.

3.26.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to ensure uniformity of size and shape.

3.26.2 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.26.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.26.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

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3.26.5 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the castings ability to perform its intended function.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Material inspection. The contractor is responsible for ensuring that supplies and materials are inspected for compliance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on the unit when a first article is required (see 3.3 and 6.2). This inspection shall include the examination of 4.3 and the tests of 4.4. The first article shall be either the first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.3 and the test of 4.4.9.

4.3 Examination. Each unit shall be examined for compliance with the requirements specified in section 3 of this document. Any redesign or modification of the contractor's standard product to

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comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.4 Tests. The first article shall receive the tests of 4.4.1 through 4.4.9. Each production unit shall receive the test of 4.4.9. Failure to pass any test shall constitute cause for rejection.

4.4.1 Performance test. The crusher shall be tested as a unit as specified in 3.1. Quarry rock shall be of the material as specified in 3.8 containing some rock 18 inches by 28 inches (457 mm by 711 mm). Only that maintenance established by the contractor and submitted as a maintenance schedule prior to the tests shall be performed during the test.

4.4.2 Percentage of crush. The flow of material shall be arranged so that all material passing through the wobbler or vibrating grizzly bars will be discharged to the stub conveyor or chute. Weight shall be not less than 100 tons (90 718 kg) of crushed material discharged from the main product conveyor. The material bypassed through the wobbler or grizzly bars shall be weighed separately and shall be not greater than 20 percent of the combined total of material from the discharge chute and the main product conveyor. Presence in the feed of more than 20 percent of material not requiring crushing shall constitute failure of the test.

4.4.3 Operating time. Crushed rock production capacity shall be computed on the basis of net operating time which shall be defined as gross operating time less allowable down time for the following:

a. Jamming of rock fragments larger than the maximum size of rock specified in 3.8.

b. Delay due to lack of material delivery to the crusher.

c. Normal machinery maintenance: lubrication, jaw setting adjustments, and takeups (down time due to any design deficiency shall not be considered normal maintenance).

d. Other conditions considered by the contracting officer to be beyond the control of the contractor.

Downtime required for maintenance other than lubrication and operating adjustments shall be considered as net operating time. Downtime caused by design deficiencies or failure of parts not classified as high mortality parts shall constitute failure of any test.

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4.4.4 Jaw settings. When the crusher is tested as a unit, the jaw settings shall be as follows:

Closed setting in inches	Time in hours
Maximum (not less than 5 inches (127 mm))	10
Minimum (not greater than 3 inches (76 mm))	10

Capacity in tph (mtph) and corresponding jaw setting shall be recorded as part of test reports.

4.4.5 Capacity as a unit. The unit shall be operated under the applicable test conditions specified in 4.4.1 through 4.4.4. Cumulative daily runs are permissible. Nonconformance to 3.8 shall constitute failure of this test.

4.4.6 Uncrushable material test. With the unit operating at capacity rate, a piece of uncrushable material such as tramp iron shall be dropped into the crusher. The piece shall be larger than the maximum jaw opening at the jaw setting being used, but not greater than 2 inches (50.8 mm) larger. Nonconformance to 3.12 shall constitute failure of this test.

4.4.7 Holdback device test. During performance tests, power shall not be cut off to motor-driven conveyors when carrying capacity load to determine conformance to 3.13.1. Failure to prevent slippage or reversal shall constitute failure of this test.

4.4.8 Lifting and tying down attachment test. When furnished, lifting and tying down attachments shall be tested to determine conformance to 3.24. The manufacturer may furnish certified engineering calculations in lieu of a test to prove that the lifting and tying down attachments offered meet the requirements set forth in MIL-STD-209.

4.4.9 Production unit operational test. Each unit produced in fulfillment of a Government contract or order shall be completely assembled, adjusted, lubricated, or otherwise serviced for operation. The motors shall be started and subjected to a warm-up period as recommended by the manufacturer. The unit shall be given a run-in test and all controls operated a sufficient number of times to ascertain that the components and mechanisms actuated by the controls operated promptly, fully, and without restriction or malfunction. Each wheeled unit shall be towed a distance of not less than 10 miles (16 km) at an average speed of not less than 15 mph (24 km/h) over unpaved road. Failure to pass any phase of this test shall be cause for rejection.

5. PACKAGING

5.1 Packaging requirements. The preservation, packing, and marking shall be as specified in the contract or order.

6. NOTES

(This section contains information of a general or explanatory nature that is helpful, but is not mandatory.)

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6.1 Intended use. The jaw crusher is intended for use as the primary rock crushing unit in a rock crushing plant.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type unit required (see 1.2)
c. When a first article is required for inspection and approval (see 3.3, 4.2.1, and 6.3).
d. When the major components require lifting eyes (see 3.6).
e. Type of feeder required (see 3.9).
f. Color of finish coat required (see 3.19)

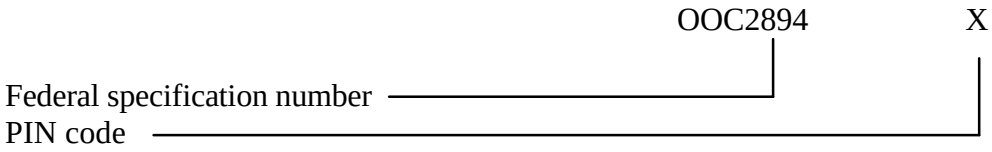
6.3 First article. When a first article inspection is required, the unit will be tested and should be a first production unit or it may be a standard production unit from the contractor's current inventory as specified in 4.2.1. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Supersession data. This specification replaces Military Specification MIL-C-29221A(YD) dated 13 September 1988.

6.5 Classification cross reference. Changes in the classification used in this specification (see 1.2), and to those found in the superseding Military Specification MIL-C-29221A(YD) are as follows:

MIL-C-29221A(YD)	OO-C-2894
Not designated	Type I
Not designated	Type II

6.6 Part or identifying numbers (PINs). The specification number and type are combined to form PINs for jaw crushers covered by this document (see 1.2). PINs are established as follows:



6.7 Class to code identifier. The class of the jaw crusher is identified by a single-digit number as shown:

TYPE	CODE NUMBER
Type I	1
Type II	2

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6.8 Subject term (key word) listing.

Crusher
Jaw crusher
Portable jaw crusher
Rock crusher

6.9 Changes from previous issue. Marginal notations are not used to identify changes with respect to the superseded document.

MILITARY INTERESTS:

Custodian:
Navy - YD1
Army - AT

Review Activities:
DLA - CS

CIVIL AGENCY COORDINATING ACTIVITY:
GSA - FSS

Preparing Activity:
Navy - YD1

(Project 3820-0022)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** OO-C-2894	**2. DOCUMENT DATE (YYMMDD)** 970414

3. DOCUMENT TITLE
CRUSHER, JAW, PORTABLE, PRIMARY AGGREGATE PLANT, MOBILE, 75-TON PER HOUR CAPACITY

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON *(if applicable)*	**7. DATE SUBMITTED** (YYMMDD)

8. PREPARING ACTIVITY

a. NAME RUDY PAMPLONA	b. TELEPHONE *Include Area Code)* (1) Commercial 805-982-5843 (2) AUTOVON 551-5843
c. ADDRESS *(Include Zip Code)* COMMANDING OFFICER, NCBC CODE 15E2D 1000 23RD AVENUE PORT HUENEME, CA 93043-4301	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** DEFENSE QUALITY AND STANDARDIZATION OFFICE 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22401-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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